



CMR COLLEGE OF ENGINEERING & TECHNOLOGY
(UGC AUTONOMOUS)

B.Tech III Semester Supplementary Examinations February-2024

Course Name: ANALOG & DIGITAL ELECTRONICS

(Common for CSE & IT)

Date: 07.02.2024 AN

Time: 3 hours

Max.Marks: 70

(Note: Assume suitable data if necessary)

PART-A

Answer all TEN questions (Compulsory)

Each question carries TWO marks.

10x2=20M

1. List the differences between ideal diode and practical diode. 2 M
2. Define transition capacitance. 2 M
3. Draw the symbols of NPN and PNP transistor. 2 M
4. Explain about the various regions in a transistor. 2 M
5. How a FET can be used as a voltage variable Resistance (VVR)? 2 M
6. Discuss octal number system. 2 M
7. Show both NAND and NOR gates are called Universal gates. 2 M
8. Analyze the steps for simplification of POS expression. 2 M
9. Illustrate applications of shift registers. 2 M
10. Define race around condition? How it can be avoided. 2 M

PART-B

Answer the following. Each question carries TEN Marks.

5x10=50M

- 11.A). Explain the V-I characteristics of Zener diode and distinguish between Avalanche and Zener Break downs. 10M

OR

11. B). Draw the circuit of bridge rectifier and explain its operation with the help of input and output waveforms. 10M

12. A). Explain the constructional details of Bipolar Junction Transistor. 10M

OR

12. B). Discuss the construction, principle of operation, characteristics and applications of UJT. 10M

13. A). Explain the operation of FET with its characteristics and explain the Different regions in transfer characteristics. 10M

OR

13. B). i) Solve the subtraction with the following unsigned binary numbers by taking the 2's complement of the subtrahend: 6M

(a) $100 - 110000$ (b) $11010 - 1101$

- ii) Construct a table for 4 -3 -2 -1 weighted code and write 9154 using this code. 4M

(P.T.O..)

14. A). Simplify the following Boolean expressions using K-map and implement them using NOR gates: 10M

$$F(W, X, Y, Z) = W'X'Y'Z' + WXY'Z' + W'X'YZ + WXYZ.$$

OR

14. B). i) Construct the logic diagram of a full subtractor using only 2-Input NAND gates. 8M
ii) Explain the properties of EX-OR gate. 2M

15. A). Design a 4-bit BCD Ripple Counter by using T-FF. 10M

OR

15. B). Explain the state reduction and state assignment in designing Sequential circuit. Consider one example in the above process. 10M
